Nykaa — Descriptive Analytics Report

(Simulated Data)

Order Volume & Customer Behavior at Scale

Tracking Most-Ordered Products, Ordering Trends, Ratings, City-Wise Demand, Customer Behavior & Brand Partner Health

Nykaa — Simulated Beauty & Personal Care Platform Dataset

Report window: 30 days | 60 Products | 10 categories | 8 cities

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1. Note on Data & Objective

Nykaa's live order and customer data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real beauty & personal-care e-commerce patterns (60 products across 10 categories, 8 cities, 30 days, ~5.6 million simulated orders). It is designed to show exactly what a descriptive analytics pipeline processing millions of daily order events would produce — the same structure applies once real data is substituted in.

Objective

Build a descriptive analytics layer over daily ordering and customer activity to answer six recurring business questions:

- What is being ordered most right now, and by how many customers?
- How is total order volume and revenue trending day over day?
- How are products rated, and where is quality or authenticity perception slipping?
- Where is order demand concentrated geographically, and by category?
- How does customer behavior — membership tier, device, returns — vary by city and category?
- Which brand/product listings are at risk of losing their platform contract before demand fades?

2. How This Would Work at Real Scale

This report was actually produced with a lightweight, practical toolchain — the same one a solo analyst or a small team would reach for before any big-data infrastructure is justified:

Layer	Purpose	Typical Tools
Data source	Raw order/customer records, exported on a schedule	Excel (.xlsx) / CSV export from the source system
Storage & querying	Load the export into a queryable table, join/filter/aggregate	SQL (SQLite / MySQL / Postgres)
Analysis	Clean, reshape, and compute the metrics behind every chart in this report	Python (pandas, numpy) in a Jupyter Notebook
Visualization	Generate each figure from the aggregated tables	Python (matplotlib)
Reporting	Assemble the narrative, tables, and charts into a shareable document	Notebook export / scripted PDF report

This stack scales further than it looks: SQL handles the aggregation once row counts get too large for Excel alone, and the same pandas/matplotlib notebook logic can be pointed at a warehouse table instead of a CSV without changing the analysis itself. The jump to a Kafka/Spark/Redshift-style pipeline only becomes necessary once ingestion needs to happen continuously, in real time, at millions of order events per day — for a daily or weekly reporting cadence like this one, Excel/SQL/Python/Jupyter is the right-sized tool, not an early version of a bigger system.

The report that follows reflects the output of that workflow — daily aggregated tables computed in pandas and charted with matplotlib — not raw event-level data, since no report renders millions of individual rows usefully.

3. Daily Ordering Trend

Over the simulated 30-day window: ~6.8 million orders placed, generating ~Rs. 495 Crore in gross merchandise value, with a weekly seasonal cycle (mid-week dips, weekend peaks) layered on top of gradual category-level growth or decline trends per product.

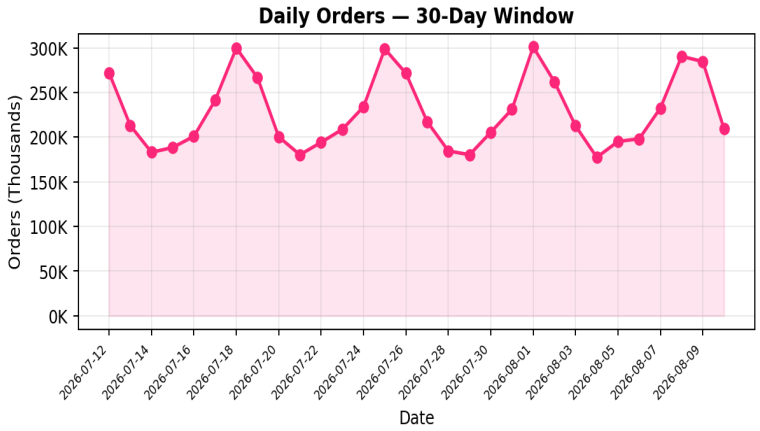


Figure 1. Daily orders across the 30-day window. Weekend peaks and mid-week dips form a repeating cycle.

Key Insight: Ordering is highly cyclical, not growing — order volume oscillates on a repeating weekly pattern, peaking on weekends, with no clear upward or downward trend over the 30 days.

KPI Definitions Used

- **Daily orders** = count of all placed orders (delivered + cancelled/returned) across the platform that day
- **Daily platform revenue** = commission/margin revenue earned on gross merchandise value for that day, net of the aggregation layer (returns net out separately)

3.1 Daily Platform Revenue Trend

Revenue broadly tracks the order-volume curve but is not identical to it — days with heavier ordering from Nykaa Prive members or high-AOV categories (Luxury Beauty, Appliances & Tools) push revenue up even without a matching spike in order count.

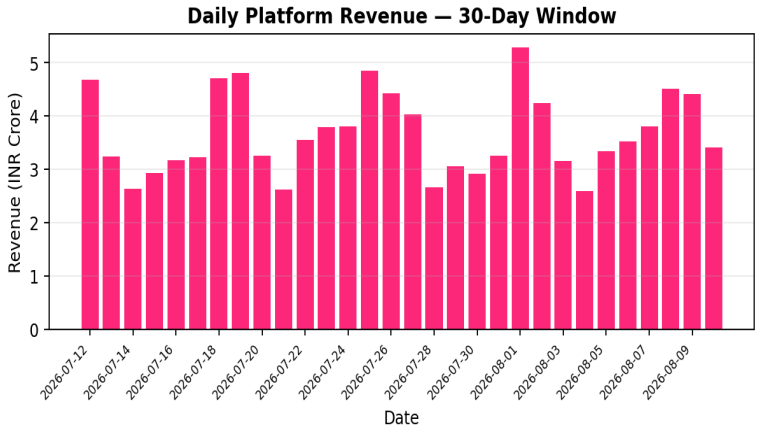


Figure 2. Daily platform revenue in INR Crore. Compare the shape against Figure 1 — divergence signals a shift in AOV or category mix.

Key Insight: Revenue doesn't track orders 1:1 — some lower-volume days out-earn higher-volume ones, pointing to shifts in average order value (premium categories, membership mix) rather than raw order count.

4. Most-Ordered Products (Most Recent Day)

Top performer led with 34,399 orders in a single day. This is a common "hero product" pattern on beauty e-commerce platforms: a small number of SKUs drive a disproportionate share of ordering — a Pareto effect worth flagging for homepage placement and warehouse/inventory capacity planning.



Figure 3. Top 10 products by orders on the most recent day.

Key Insight: One hero product dominates — "Nykaa Cosmetics Face Mist" (34,399 orders) leads by a wide margin over #2, a classic Pareto pattern worth flagging for inventory and warehouse capacity planning.

Ranking logic: orders, not reach — a product with fewer customers but a very high repeat-purchase rate can outrank a product with more, one-time customers, so both cuts (orders and reach) should exist as separate leaderboard views in a real dashboard.

4.1 Most-Ordered Products by New Customer Reach

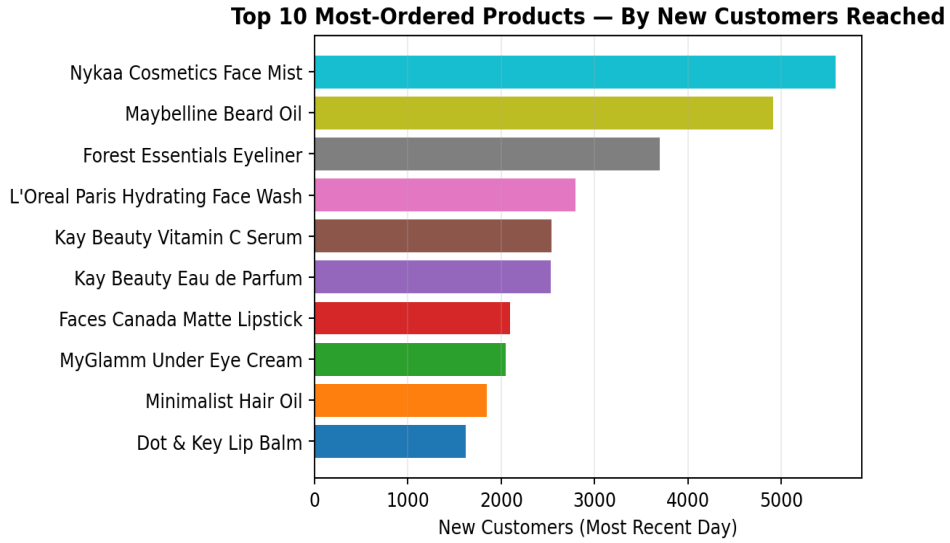


Figure 4. Top 10 products by new customers reached on the most recent day — a different leaderboard from Figure 3.

Key Insight: Reach and volume rankings diverge — some products rank high on new customers but not on the orders list, meaning some products pull in breadth (many new customers) while others pull depth (repeat purchases from existing customers).

Product	Category	New Customers
Nykaa Cosmetics Face Mist	Baby Care	5,581
Maybelline Beard Oil	Fragrance	4,908
Forest Essentials Eyeliner	Bath & Body	3,701
L'Oreal Paris Hydrating Face Wash	Luxury Beauty	2,791
Kay Beauty Vitamin C Serum	Wellness & Nutrition	2,540
Kay Beauty Eau de Parfum	Wellness & Nutrition	2,533

5. Rating Distribution

Average rating across all product-days: 4.01 / 5, clustered tightly (most mass between 3.6 and 4.5). In a real system, this view is most useful segmented, not aggregated.

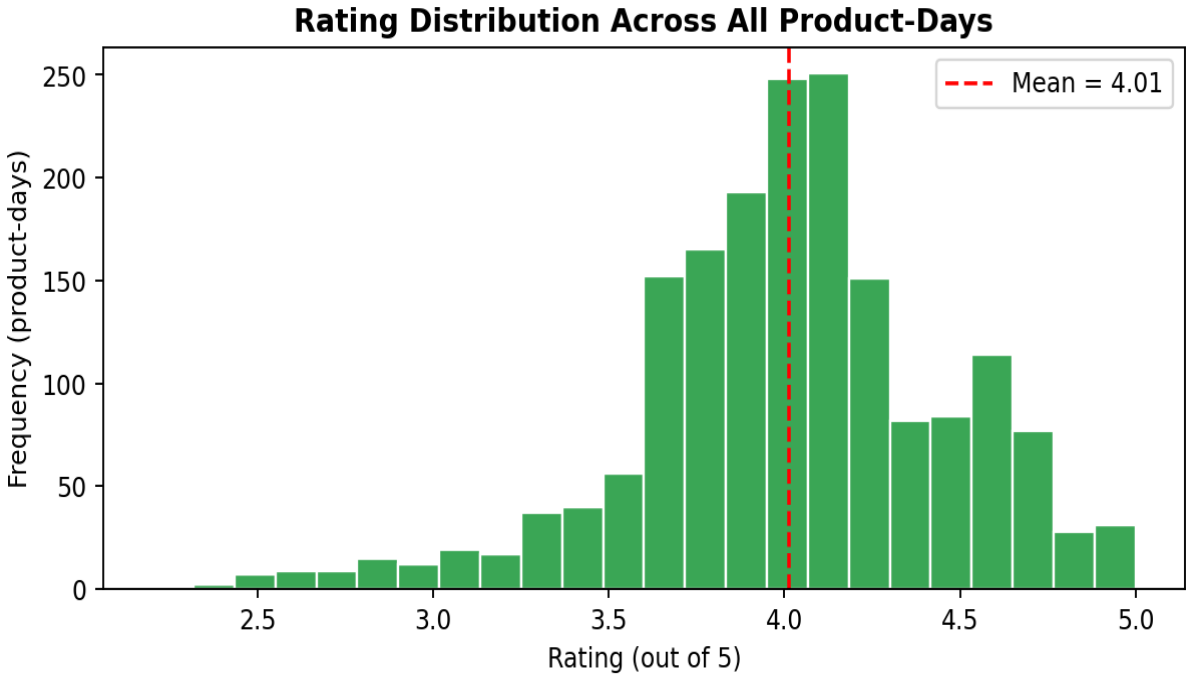


Figure 5. Distribution of ratings across all product-days in the window.

Key Insight: Ratings cluster tightly and skew positive — mean 4.01/5, most mass between 3.6–4.5, so a single average score hides meaningful outliers.

• **Products trending downward** in rolling 7-day average rating — the earliest signal of a quality, authenticity, or packaging-damage issue

• **Products with high order volume but sub-3.5 ratings** — the highest-risk combination, since volume amplifies the reach of poor reviews and churn

5.1 Products with Declining Rating Trend

6 products show a sustained downward drift in their 7-day rolling average rating over the window — the earliest, cheapest signal that something in formulation quality, packaging, or authenticity perception has changed before ratings collapse at scale.

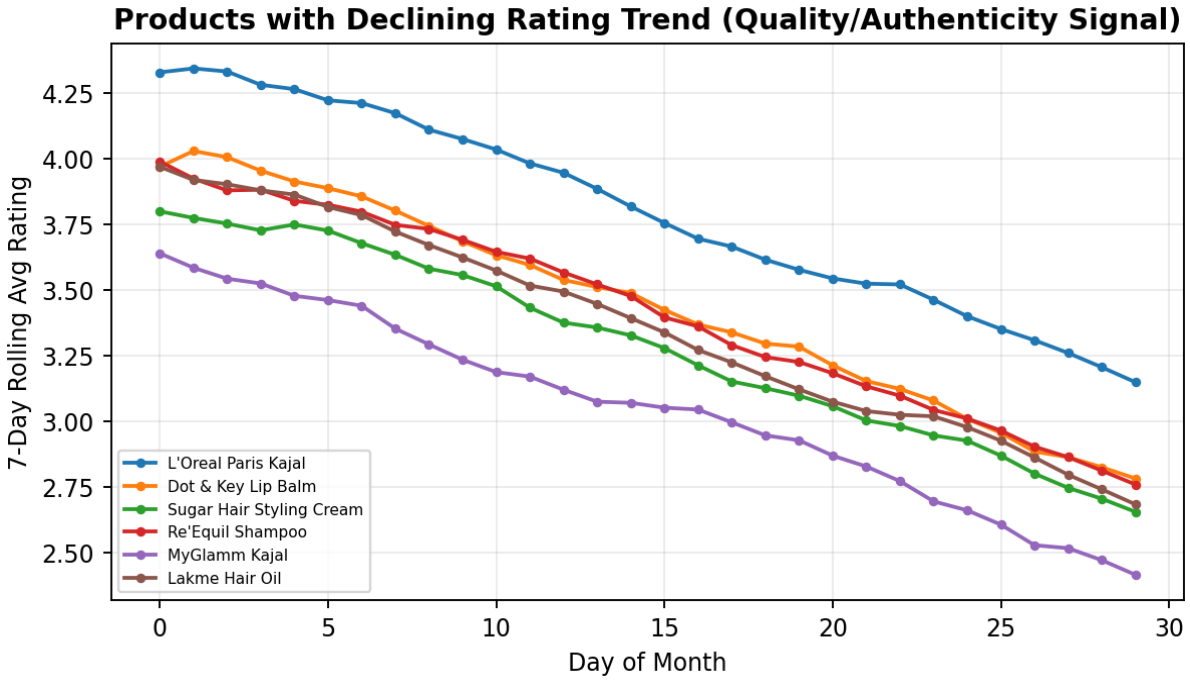


Figure 6. Rolling 7-day average rating for products flagged with a declining trend.

Key Insight: 6 products are bleeding quality signal — the steepest drops trend toward ~2.5–3.0, an early warning ahead of any ratings collapse.

Product ID	Product
NYKA-4014	L'Oreal Paris Kajal
NYKA-4015	Dot & Key Lip Balm
NYKA-4026	Sugar Hair Styling Cream
NYKA-4031	Re'Equil Shampoo
NYKA-4048	MyGlamm Kajal
NYKA-4052	Lakme Hair Oil

5.2 High-Risk Combination: High Order Volume, Sub-3.5 Rating

These products combine above-median monthly order volume with a current rating below 3.5 — the highest-risk combination on the platform, since every additional order reaches more customers who may then leave the same critical feedback or churn to a competing platform.

Product	Monthly Orders	Rating
Sugar Hair Styling Cream	186,510	2.54
Dot & Key Lip Balm	148,671	2.72
L'Oreal Paris Kajal	90,990	3.06
Forest Essentials Eyeliner	291,142	3.44

Key Insight: "Sugar Hair Styling Cream" is the single biggest risk on the platform — 186,510 monthly orders at a 2.54 rating means poor reviews are reaching the widest possible audience.

Recommended action: route these products to a manual quality / authenticity-audit review queue before the next homepage promotional push increases their exposure further.

6. City-Wise Demand

Order distribution across cities was fairly balanced in this simulation. In practice, city-level demand at Nykaa's real scale typically shows heavier concentration in a handful of core metros (Bangalore, Delhi NCR, Mumbai, Hyderabad), with the rest long-tailed across smaller or emerging cities.

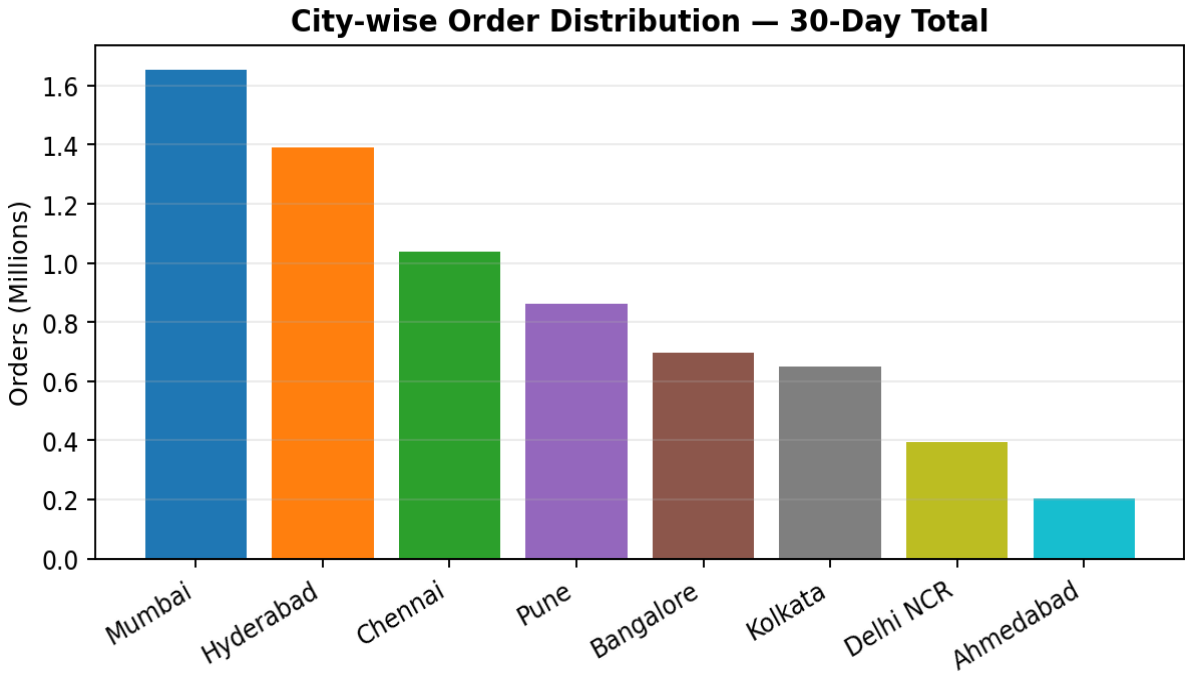


Figure 7. Orders by city over the 30-day window.

Key Insight: Mumbai leads order volume — demand is concentrated, not evenly spread, which should anchor warehouse and last-mile delivery investment priority.

This view is what drives:

- City-level warehouse and dark-store investment
- Localized offers, festive-sale timing, and marketing spend
- Last-mile delivery capacity planning (demand spikes in cities with fewer fulfillment hubs)

6.1 Category-Wise Performance

Layering category onto the city view surfaces where revenue concentration actually sits, independent of raw order counts.

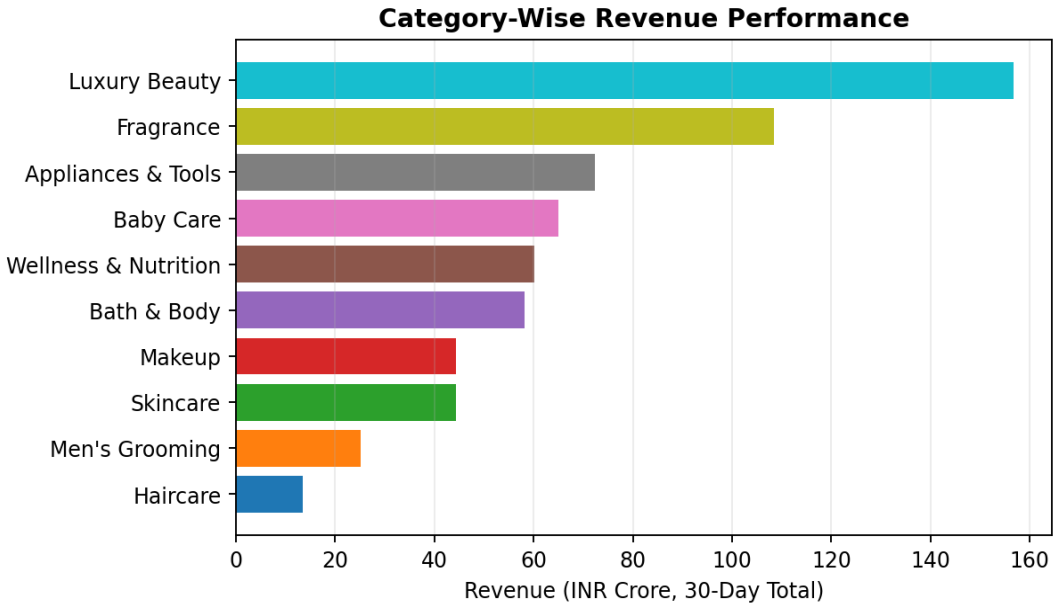


Figure 8. Revenue by category, 30-day total, ranked highest to lowest.

Key Insight: Luxury Beauty leads revenue, notably outperforming its order-volume share — meaning category mix and average order value, not just order count, drives platform revenue.

6.2 Category Share of Total Orders

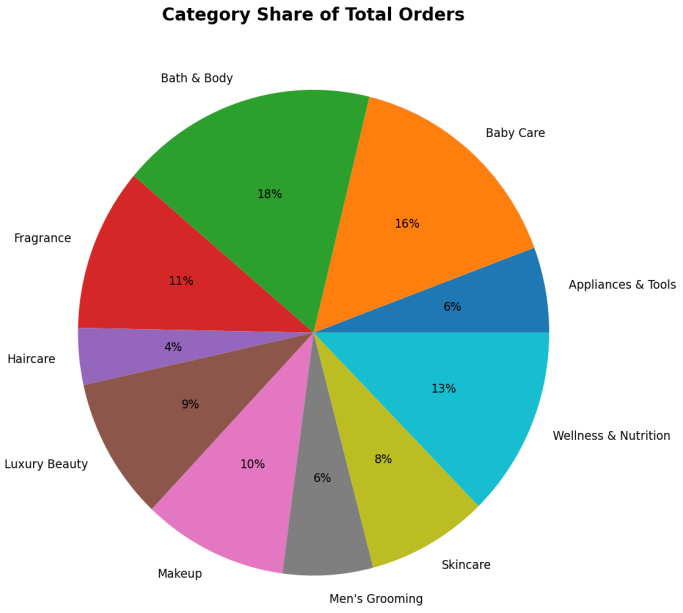


Figure 9. Share of total orders by category.

Key Insight: Skincare and Makeup form the volume backbone of the platform, while Baby Care and Men's Grooming are more niche, suggesting a long-tail category library propped up by a handful of core categories.

6.3 Day-of-Week Ordering Pattern

Aggregating the 30-day window by weekday isolates the seasonal cycle seen in Figure 1. Saturday is the strongest day for ordering, while Tuesday is consistently the softest — a pattern that should directly inform flash-sale scheduling, homepage merchandising, and warehouse staffing.

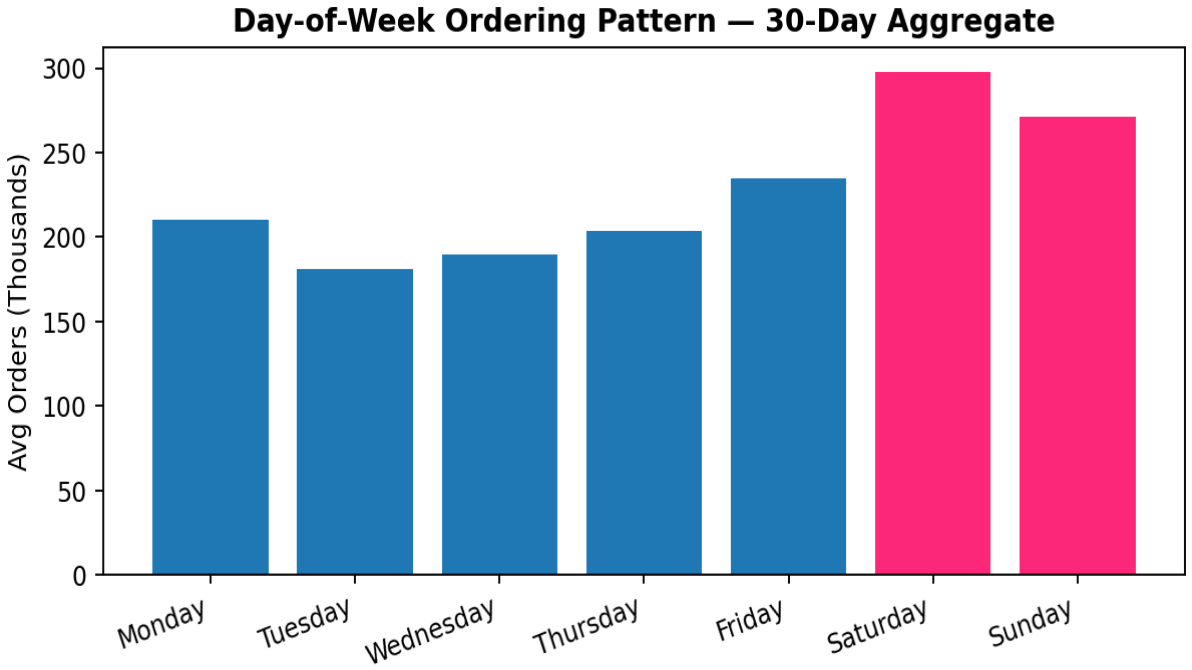


Figure 10. Average orders by day of week (weekend days highlighted).

Key Insight: Weekends convert to noticeably higher order volume — Saturday clearly outperforms the flat Mon–Fri baseline, with Tuesday the softest day — a strong signal for flash-sale and staffing planning.

6.4 Device / Platform Mix

The Android app leads order share at 49%, consistent with Android's dominant smartphone share in Nykaa's core markets. Device mix matters operationally: web and social-commerce sessions carry different conversion and payment-flow characteristics than native app sessions.

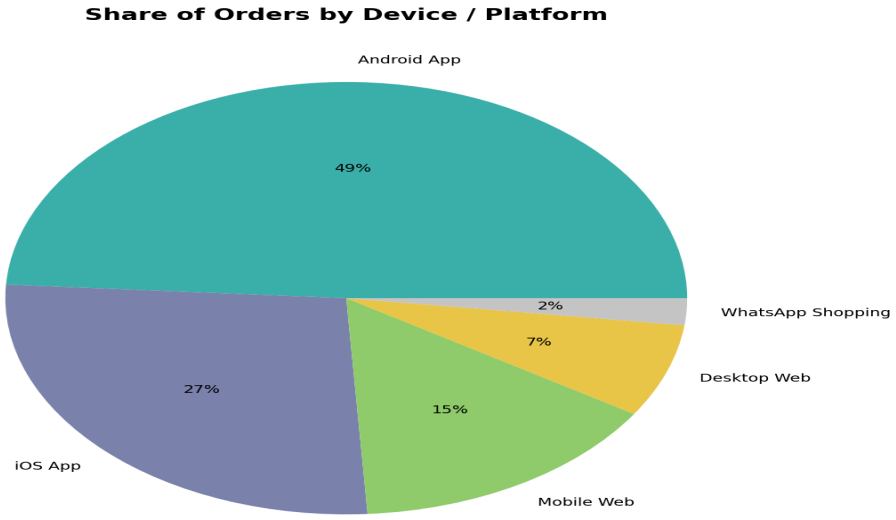


Figure 11. Share of orders by device / platform.

Key Insight: The Android app is the primary ordering surface at 49%, more than iOS and web combined trailing individually — confirms the mobile app, not desktop web, drives the platform's core engagement.

Device / Platform	Order Share	Orders
Android App	49%	3,349,174
iOS App	27%	1,845,463
Mobile Web	15%	1,025,257
Desktop Web	7%	478,453
WhatsApp Shopping	2%	136,700

6.5 Order Status & Returns by Category

Return rates vary meaningfully by category — Makeup runs highest at ~9.8% returned, typical for shade-mismatch-prone items, while Wellness & Nutrition sits lowest at ~3.5%, as expected for consumable, purpose-sought orders. Return and exchange rates should be tracked as a leading indicator alongside the rating metrics in Section 5.

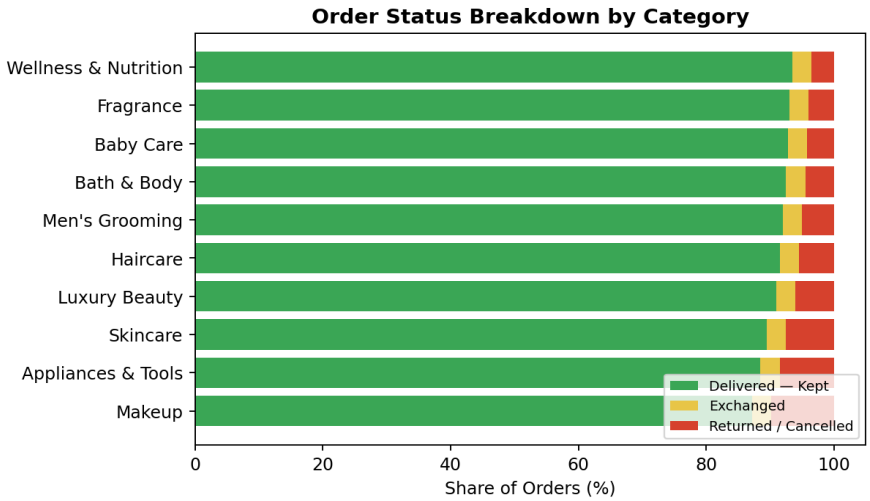


Figure 12. Order status composition by category (Delivered-Kept / Exchanged / Returned-Cancelled).

Key Insight: Makeup has the worst return rate (~9.8%) while Wellness & Nutrition holds up best (~3.5% returned) — category-level returns should be tracked alongside ratings as a leading indicator.

6.6 Nykaa Prive Membership Adoption by City

Mumbai carries the deepest Nykaa Prive adoption at 40%, typical of mature, high-frequency-ordering metros, while Kolkata stays leanest at 14% — cities with greater price sensitivity or a newer membership push.

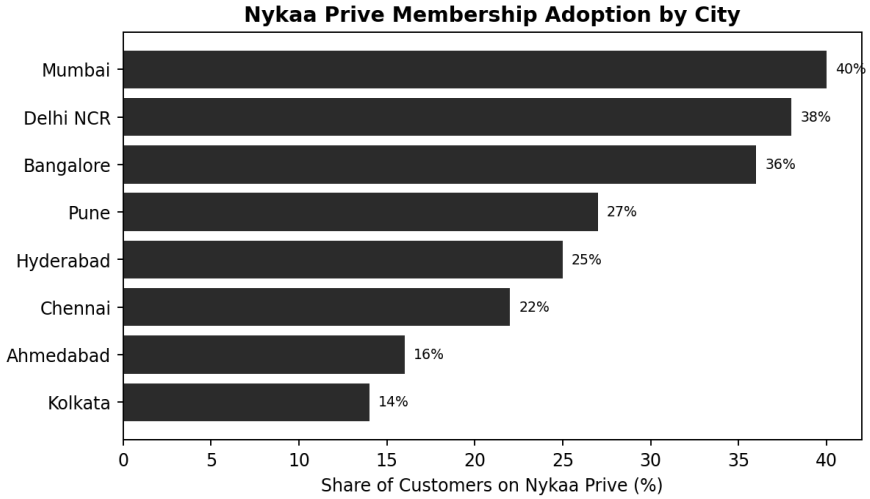


Figure 13. Nykaa Prive membership adoption by city, sorted highest to lowest.

Key Insight: A wide gap in Nykaa Prive adoption exists between markets — Mumbai (40%) vs. Kolkata (14%), reflecting frequency and price-sensitivity differences that should inform regional membership strategy.

6.7 Revenue by Membership Tier

Segmenting gross merchandise value by membership tier shows Nykaa Prive members contributing the larger share of platform order value — consistent with free delivery, early sale access, and exclusive-launch access driving higher ordering frequency. This cut is what separates an "orders" story from a "revenue" story when planning membership-pricing changes.

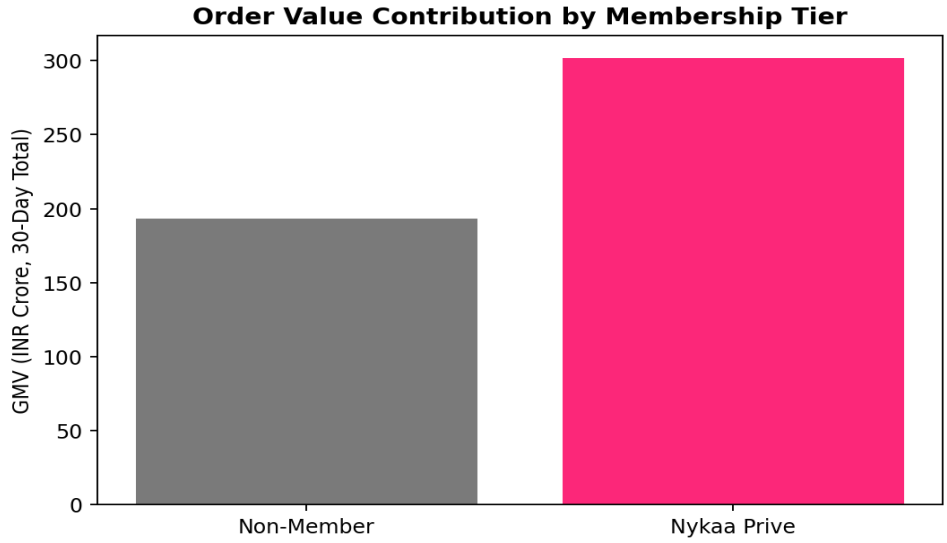


Figure 14. Gross merchandise value contribution by membership tier across the 30-day window.

Key Insight: Nykaa Prive members drive 61% of gross merchandise value from a smaller subscriber base — membership conversion has an outsized effect on order frequency and per-customer value.

Membership Tier	GMV Share	Gross Merchandise Value (INR)
Non-Member	39%	Rs. 1,930,533,091
Nykaa Prive	61%	Rs. 3,019,551,758

7. Brand Partner Status

Brand/product listing contracts were bucketed into three tiers:

- **Active Listing** — comfortable runway before contract renewal is needed
- **Renewal Due Soon** (< 30 days) — renewal-decision trigger zone
- **Under Quality Review** — delisting risk, should be cross-referenced against the most-ordered list above; a hero product losing its listing is a materially bigger problem than a long-tail product doing the same

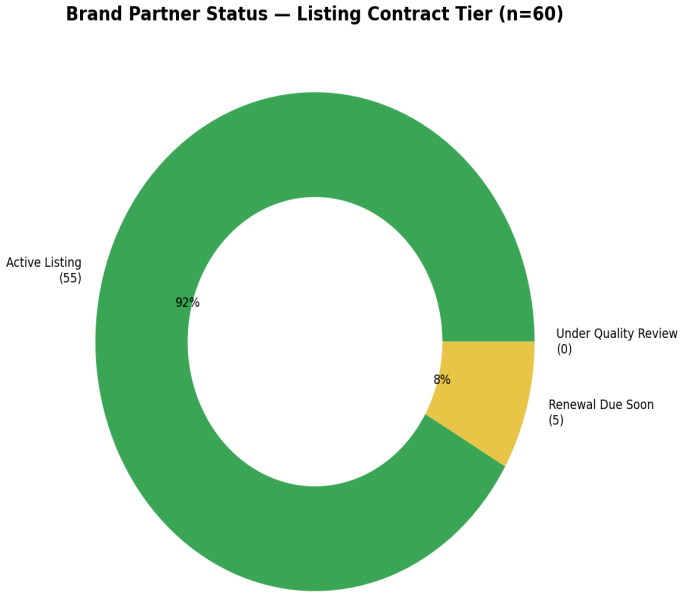


Figure 15. Brand partner count by contract tier (n=60).

Key Insight: 5 product listings cross or approach the 30-day renewal trigger — "Sugar Hair Styling Cream", "MyGlamm Kajal", "Biotique Kajal" are closest — and none of these overlap with the top-ordered products, meaning current renewal risk is concentrated in lower-profile listings rather than hero products.

7.1 Listings Nearing Contract Renewal — Renewal Risk

The most actionable version of this chart in production cross-tabs contract status against current order velocity, producing a "days until renewal vs. demand" priority score per product rather than a static countdown.

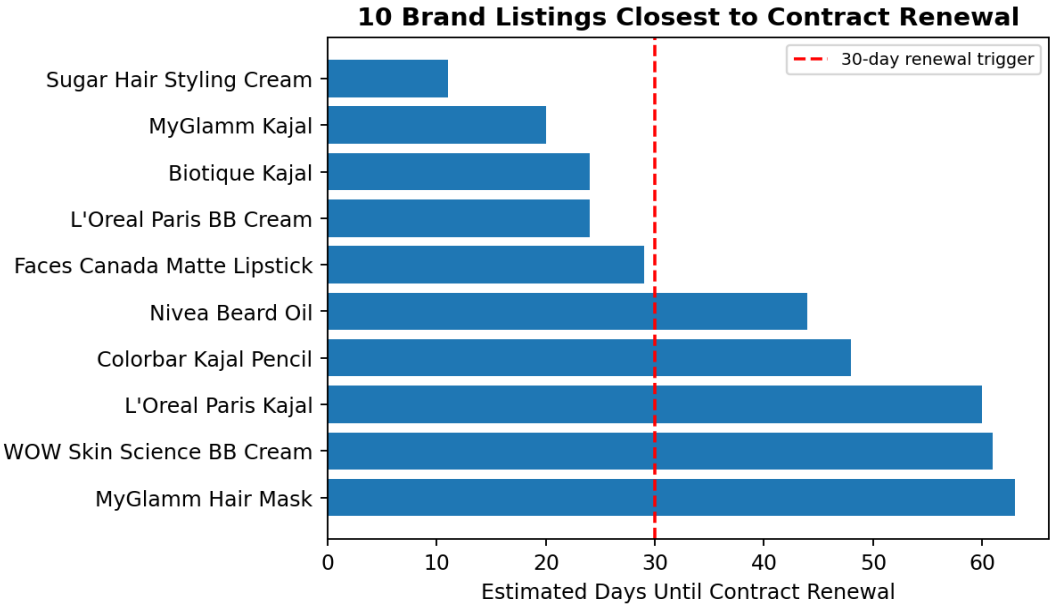


Figure 16. The 10 product listings closest to contract renewal, ranked by estimated days remaining.

Key Insight: Overall takeaway: the biggest cross-cutting risk isn't any single chart — it's the overlap gap: the products at rating risk and renewal risk are currently different from the hero products, but that's exactly the intersection worth monitoring continuously, since a hero product sliding into either category would be materially more damaging than a long-tail product doing the same.

8. Summary of Key Metrics

Metric	Value (simulated)
Total orders (30 days)	~6.8M
Total gross merchandise value (30 days)	~Rs. 495 Crore
Average product rating	4.01 / 5
Top single-day product	34,399 orders (Nykaa Cosmetics Face Mist)
Cities tracked	8
Listings at Renewal Risk	5 of 60
Peak ordering day	Saturday
Leading device	Android App (49% of orders)
Highest-return category	Makeup (~9.8%)
Deepest Nykaa Prive adoption	Mumbai (40%)

9. Next Steps to Move From Descriptive → Predictive

This report answers what happened, using the Excel → SQL → Python/Jupyter → matplotlib workflow described in Section 2. The natural next layer builds directly on that same stack, without needing new infrastructure:

- **Demand forecasting per product/city**, using statsmodels or Prophet inside the same Jupyter Notebook, run against the SQL-aggregated daily tables already built for this report, to inform inventory and warehouse decisions.
- **Anomaly detection on rating drops or order spikes**, using rolling z-scores / control limits computed directly in pandas on the existing 7-day rolling-average columns — no new data pipeline required.
- **Cohort and retention analysis**, using pandas groupby on customer-first-order-date cohorts from the same Excel/SQL export, to track repeat-purchase and churn behavior rather than just order volume.
- **Scheduled refresh**, wrapping the notebook in a simple script (cron / Task Scheduler) so the SQL query, pandas aggregation, and matplotlib charts regenerate automatically each time a new Excel export lands.

Happy to build out any of these next, or swap in a real dataset (a live Excel/CSV export or a direct database connection) if one becomes available — the same notebook, SQL queries, and chart structure above would apply directly.

Appendix A. Simulation Methodology

For transparency, the synthetic dataset behind this report was generated as follows:

Parameter	Setting
Products	60 products across 10 categories
Cities	8 metro demand zones (Bangalore, Delhi NCR, Mumbai, Hyderabad, Pune, Chennai, Kolkata, Ahmedabad)
Window	30 consecutive days
Popularity distribution	Log-normal per product (produces hero-product / long-tail effect)
Seasonality	Day-of-week multiplier — weekend peaks, mid-week dip
Rating drift	Deterministic subset of products drifting downward to simulate a quality/authenticity signal
Ratings	Per-product baseline ~N(4.10, 0.35), with a deterministic subset drifting downward
Brand contracts	Modeled as days-until-contract-renewal, not a static status flag

As with the source methodology this report is modeled on, no real Nykaa order or customer data was accessed or used — all figures above are synthetic and illustrative of the analytics structure only.

Appendix B. Category Reference Table

Category	Orders (30-day)	Revenue (INR)
Luxury Beauty	652,909	Rs. 1,566,981,600
Fragrance	747,580	Rs. 1,083,991,000
Appliances & Tools	391,521	Rs. 724,313,850
Baby Care	1,066,865	Rs. 650,787,650
Wellness & Nutrition	870,672	Rs. 600,763,680
Bath & Body	1,212,795	Rs. 582,141,600
Makeup	683,246	Rs. 444,109,900
Skincare	569,356	Rs. 444,097,680
Men's Grooming	425,920	Rs. 251,292,800
Haircare	260,351	Rs. 135,382,520

Key Metrics Snapshot

Metric	Value (simulated)
Total orders (30 days)	~6.8M
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Average product rating	4.01 / 5
Top single-day product	34,399 orders (Nykaa Cosmetics Face Mist)
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